Dark Energy as Residual Vacuum Free Energy: A Thermodynamic Bound on the Cosmological Constant

Part of the FST Programme [35]

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Abstract

This revision audits the mathematical core of the proposed thermodynamic dark-energy framework. For the dimensionally closed reduced function $\Phi_{\mathrm{red}} = A + B\rho\ln(\rho/\rho_{\mathrm{Pl}}) + C\rho^2$, $B, C > 0$, existence, uniqueness, strict convexity, and an explicit Lambert-W minimiser are proven. The original momentum-space expression is dimensionally inhomogeneous and does not determine the matching coefficients. Consequently the scaling $\Lambda_{\mathrm{eff}} \sim H_0^2/[c^2\ln(M_{\mathrm{Pl}}/H_0)]$ remains an external RG-matching ansatz. Its corrected numerical value is $3.8\times10^{-55}\,\mathrm{m}^{-2}$, about 290 below observation.

The scalar-sector audit also changes the physical status. The Pöschl–Teller potential $V_0\,\mathrm{sech}^2(\phi/\phi_0)$ has an unstable maximum at $\phi = 0$; the earlier tanh history is a phenomenological proxy, not a derived stable attractor. Standard quasistatic metric $f(R)$ perturbations approach a slip of 1/2 (or reciprocal 2, by convention) and a GR-like lensing combination $\Sigma \simeq 1/F$, not 5/4 and 9/8. Finally, the Hu–Sawicki environment ledger yields correct necessary curvature thresholds but no source-bound local profile, so neither $|f_{R0}| = 1.2\times10^{-5}$ nor 10^{-6} is by itself a Cassini proof. The framework is therefore a conditional research programme, not a closed solution of the cosmological-constant or screening problems. **Keywords:** cosmological constant,

dark energy, vacuum energy, scalar-tensor gravity, gravitational slip

1 Introduction

This revision records the mathematical status after a line-by-line proof/paper audit on 16 July 2026. The purpose is narrower than that of the earlier skeleton: it separates the statements that follow from a dimensionally closed one-variable model from the physical identifications that remain open.

The audit changes four headline claims. First, the original momentum-space integrand was dimensionally inhomogeneous and therefore did not define a physical free-energy density without explicit matching coefficients. Second, the proposed Pöschl–Teller saturation law was not derived from the canonical Klein–Gordon–Friedmann system and its hilltop is unstable. Third, the thermodynamic beta function defined as the derivative of a stationarity condition is identically zero. Fourth, standard metric $f(R)$ perturbation theory does not predict the previously stated values $\eta = 5/4$ and $\Sigma = 9/8$. These claims are withdrawn below.

2 Dimensionally closed variational model

2.1 Raw motivation and its limitation

The earlier motivating expression was

$$\Phi_{\mathrm{raw}}(\rho) = \int_{\Lambda_{\mathrm{IR}}}^{\Lambda_{\mathrm{UV}}} \left[\frac{\omega(k)}{2} + T_{\mathrm{dS}}\rho \ln\left(\frac{\rho}{\rho_{\mathrm{Pl}}}\right) + \frac{8\pi G\rho^2}{k^2}\right] \frac{4\pi k^2}{(2\pi)^3}\,\mathrm{d}k. \tag{1}$$

In natural units, the three terms inside the square brackets have dimensions E, E^5, and E^4, respectively. They cannot be added as written. Planck-unit rescaling does not repair the problem: it hides powers of M_{Pl} and H, precisely the powers that determine the proposed cosmological scaling. Equation (1) is therefore only a schematic motivation, not a physical functional and not a derivation from a mode sum.

2.2 Reduced model

Definition 2.1 (Reduced free-energy density)**.** For $\rho > 0$, let

$$\Phi_{\mathrm{red}}(\rho) = A + B\,\rho \ln\left(\frac{\rho}{\rho_{\mathrm{Pl}}}\right) + C\rho^2, \qquad B > 0, \quad C > 0. \tag{2}$$

Here $[\rho] = [\Phi_{\mathrm{red}}] = E^4$, $[A] = E^4$, $[B] = 1$, and $[C] = E^{-4}$. The constants B and C are renormalised matching coefficients. Their values are not supplied by Eq. (1).

The kernel

$$V_{\mathrm{grav}}(\rho, k) = \frac{8\pi G\rho^2}{k^2} \tag{3}$$

has the dimension of an energy density and can motivate a positive quadratic term after an explicitly specified integration and matching prescription. Dimensional analysis selects it as a lowest-order monomial only after the assumptions of one power of G, quadratic dependence on ρ, and inverse powers of k are imposed. It does not establish uniqueness among general kernels, and a quasistatic 4/3 force enhancement does not by itself derive an off-shell free-energy functional.

3 Proven result and conditional scaling

Theorem 3.1 (Unique minimiser of the reduced model)**.** *The function* $\Phi_{\rm red}$ *in Definition 2.1 has exactly one global minimiser* $\rho_* > 0$*. It satisfies*

$$B\left[1+\ln\left(\frac{\rho_*}{\rho_{\rm Pl}}\right)\right]+2C\rho_*=0 \tag{4}$$

and is given explicitly by

$$\rho_*=\frac{B}{2C}\,W\!\left(\frac{2C\rho_{\rm Pl}}{eB}\right), \tag{5}$$

where W *is the principal Lambert function.*

Proof. The first derivative tends to $-\infty$ as $\rho\to 0^+$ and to $+\infty$ as $\rho\to\infty$. Moreover,

$$\Phi''_{\rm red}(\rho)=\frac{B}{\rho}+2C>0. \tag{6}$$

Thus the derivative has exactly one zero and strict convexity makes that zero the unique global minimiser. Rearranging Eq. (4) gives $(2C\rho_*/B)\exp(2C\rho_*/B)=2C\rho_{\rm Pl}/(eB)$, which yields Eq. (5). □

The associated curvature convention is

$$\Lambda_{\rm eff}=\frac{8\pi G}{c^4}\rho_*. \tag{7}$$

No dependence on H_0 follows until B/C is independently matched. The frequently used expression

$$\Lambda_{\rm eff}^{\rm ansatz}\sim\frac{H_0^2}{c^2\ln(M_{\rm Pl}/H_0)} \tag{8}$$

is therefore a conditional RG-matching ansatz, not a consequence of Theorem 3.1.

Corollary 3.2 (Arithmetic of the conditional ansatz)**.** *For* $H_0=67.4\,{\rm km\,s^{-1}\,Mpc^{-1}}$ *and* $L=\ln(M_{\rm Pl}/H_0)\simeq 140$*,*

$$\frac{H_0^2}{c^2L}\simeq 3.8\times10^{-55}\ {\rm m^{-2}}. \tag{9}$$

Compared with $\Lambda_{\rm obs}\simeq 1.11\times10^{-52}\ {\rm m^{-2}}$ *from Planck 2018 [4], the ansatz is low by a factor of about* 2.9×10^2*. This is an unresolved normalisation/matching factor, not order-of-magnitude agreement.*

3.1 Scalar-sector status

Consider the phenomenological Lagrangian

$$\mathcal{L}=\frac{R+\gamma R^2}{16\pi G}+\frac{1}{2}(\partial\phi)^2-V_{\rm PT}(\phi)+\mathcal{L}_m,\qquad V_{\rm PT}(\phi)=V_0\,{\rm sech}^2(\phi/\phi_0). \tag{10}$$

For the quadratic curvature sector the scalaron mass is $m_s^2=1/(6\gamma)$. The scalar potential obeys

$$V'_{\rm PT}(0)=0,\qquad V''_{\rm PT}(0)=-\frac{2V_0}{\phi_0^2}<0. \tag{11}$$

Hence $\phi = 0$ is a maximum, not a minimum. Linear perturbations satisfy

$$r^2 + 3Hr - \frac{2V_0}{\phi_0^2} = 0,$$

and the larger root is positive for every $V_0 > 0$. Hubble friction can delay the roll but cannot make the hilltop asymptotically stable. The earlier stable-de-Sitter theorem and oscillation forecast are withdrawn.

For a canonical scalar and pressureless matter, define

$$x = \frac{\dot{\phi}}{\sqrt{6}\,HM_{\mathrm{Pl}}}, \qquad y = \frac{\sqrt{V}}{\sqrt{3}\,HM_{\mathrm{Pl}}}, \qquad \lambda = -M_{\mathrm{Pl}}\frac{V'}{V}, \qquad N = \ln a.$$

The correct autonomous system is

$$x' = -3x + \sqrt{\frac{3}{2}}\lambda y^2 + \frac{3}{2}x(1 + x^2 - y^2), \tag{12}$$

$$y' = -\sqrt{\frac{3}{2}}\lambda xy + \frac{3}{2}y(1 + x^2 - y^2). \tag{13}$$

With $\Omega_\phi = x^2 + y^2$ and $w_\phi = (x^2 - y^2)/(x^2 + y^2)$,

$$\frac{\mathrm{d}\Omega_\phi}{\mathrm{d}N} = -3w_\phi\,\Omega_\phi(1 - \Omega_\phi). \tag{14}$$

This does not reduce to the tanh equation printed in earlier versions.

A bounded tanh curve may still be used as a data parametrisation if it is labelled as such. A dimensionally consistent non-negative choice is

$$X_{\mathrm{DE}}^{\mathrm{proxy}}(a) = \frac{\Phi_0}{2}\left[1 + \tanh\!\Big(\kappa(a - a_t)\Big)\right], \tag{15}$$

$$\frac{\mathrm{d}X_{\mathrm{DE}}^{\mathrm{proxy}}}{\mathrm{d}a} = 2\kappa X_{\mathrm{DE}}^{\mathrm{proxy}}\left(1 - \frac{X_{\mathrm{DE}}^{\mathrm{proxy}}}{\Phi_0}\right). \tag{16}$$

Here $X_{\mathrm{DE}} = \rho_{\mathrm{DE}}/\rho_{\mathrm{crit},0}$, not the instantaneous fraction $\rho_{\mathrm{DE}}/\rho_{\mathrm{crit}}(a)$. Equations (15)–(16) are a phenomenological proxy and are not derived from Eq. (10).

4 Kinematic consequences of a density proxy

The conditional estimate from Eq. (8) is $3.8\times10^{-55}\,\mathrm{m}^{-2}$, not $3.8\times10^{-53}\,\mathrm{m}^{-2}$. It is therefore about 290 times below Λ_{obs}, not a factor of three.

A bounded curve may be used as a phenomenological density history only if its normalisation is defined consistently. Let $X_{\mathrm{DE}}(a) = \rho_{\mathrm{DE}}(a)/\rho_{\mathrm{crit},0}$ and

$$E^2(a) = \frac{H^2}{H_0^2} = \Omega_m a^{-3} + X_{\mathrm{DE}}(a).$$

Then

$$q(a) = -1 - \frac{a}{2E^2}\frac{\mathrm{d}E^2}{\mathrm{d}a} = \frac{\frac{1}{2}\Omega_m a^{-3} - X_{\mathrm{DE}} - \frac{a}{2}X'_{\mathrm{DE}}}{E^2}, \tag{17}$$

$$w_{\mathrm{DE}}(a) = -1 - \frac{1}{3}\frac{\mathrm{d}\ln X_{\mathrm{DE}}}{\mathrm{d}\ln a}. \tag{18}$$

The earlier deceleration formula used the wrong coefficient for $aX'_{\rm DE}$. It also alternated between an instantaneous density fraction and a density normalised to today's critical density. Consequently, the reported acceleration redshift, w_0, CPL map, and proxy χ^2 are withdrawn pending recomputation from a physically derived history. No official DESI likelihood evaluation is claimed.

Remark 4.1 (DESI DR2 phantom crossing and scalar-tensor reconstructions). Recent analyses of DESI DR2 data in combination with Planck 2018 and Supernovae Ia suggest an effective dark energy equation of state $w(z)$ that crosses the phantom divide $w = -1$. While minimally coupled single scalar fields (quintessence) cannot cross $w = -1$ without ghost degrees of freedom, reconstructed scalar-tensor gravity theories [36] and interacting dark sector models with field-dependent dark matter masses [37, 38] naturally generate an apparent phantom divide $w_{\rm eff}(z) < -1$. The phenomenological proxy in Eq. (15) serves as a kinematic testbed for such effective scalar-tensor evolution.

5 Local consistency gates

5.1 Radiation-era trace

For an ideal perfect fluid with the sign convention used here,

$$T = (1 - 3w)\rho.$$

Thus $T = 0$ for exactly massless radiation with $w = 1/3$. This removes the leading classical trace source, but it is not a proof of BBN safety: massive species, trace anomalies, background evolution, and initial conditions must be propagated through a quantitative abundance calculation.

5.2 Convexity and screening

A fixed strictly convex function of ρ has one minimiser. That elementary fact does not constitute a no-go theorem for chameleon screening. In a chameleon model the effective scalar potential depends on ambient density, so the function being minimised changes with the environment. The previous Γ-convergence and energy-dissipation block did not supply a valid liminf/recovery proof and mixed gradients of different variables; it is withdrawn. The accompanying numerical phase plot remains, at most, a toy illustration and not a Cassini calculation.

5.3 Scope of the variational statement

The variational theorem below concerns only the reduced scalar function with specified positive coefficients. It does not solve the radiative-stability problem, determine a vacuum subtraction scheme, construct an RG trajectory, or derive a local screening profile. Those are independent gates. In particular, neither a resolvent analogy nor a metric on configuration space changes the zero set of the differential of a fixed scalar functional.

6 Renormalisation, beta response, and observational status

The physical scaling problem is concentrated in the matching coefficients B and C. A valid derivation must provide a source- and scheme-controlled ledger

$$\text{bare} \longrightarrow \text{Minkowski subtraction} \longrightarrow \text{counterterms} \longrightarrow \text{horizon remainder.}$$

No such derivation is supplied here. The RG, data-processing, one-loop, BBN, and asymptotic-safety passages in the earlier skeleton were analogies or leading-order observations, not proofs. In particular, the ideal-radiation identity $T = (1-3w)\rho = 0$ is correct for $w = 1/3$, but it does not by itself guarantee sub-percent BBN safety once massive species, trace anomalies, and the background scalar dynamics are included.

The earlier definition

$$\beta_\Lambda^{\rm old} = \frac{\mathrm{d}}{\mathrm{d}\ln a} \left.\frac{\partial\Phi}{\partial\rho}\right|_{\rho=\rho_*(a)}$$

is identically zero because the bracket is zero for every a. It cannot yield a non-zero flow. If one merely differentiates the conditional ansatz $\rho_*^{\rm ansatz} \propto H^2 M_{\rm Pl}^2/L$, $L = \ln(M_{\rm Pl}/H)$, the result is

$$\frac{\mathrm{d}\rho_*^{\rm ansatz}}{\mathrm{d}\ln a} = \rho_*^{\rm ansatz}\left(2 + \frac{1}{L}\right)\frac{\mathrm{d}\ln H}{\mathrm{d}\ln a}. \tag{19}$$

This is a response of an assumed scaling, not an independently extracted beta function and not evidence for that scaling.

No Wilsonian beta function, fixed-point monotonicity, bound on γ at one loop, or Kullback–Leibler data-processing statement is promoted to a result in this revision. Each requires a dimensionally complete action, normalised probability objects where information inequalities are invoked, and an explicit renormalisation prescription.

The conditional scale

$$\Lambda_{\rm eff}^{\rm ansatz} \simeq 3.8 \times 10^{-55}\ \mathrm{m}^{-2}$$

is about 290 times below the observed value. Accordingly, this paper does not claim a solution of the cosmological-constant or coincidence problems, a DESI-compatible equation of state, or a stable late-time attractor.

6.1 Metric $f(R)$ perturbations

Use the convention

$$\mathrm{d}s^2 = -(1+2\Psi)\mathrm{d}t^2 + a^2(1-2\Phi)\mathrm{d}\mathbf{x}^2$$

and define $s(k) = (k^2/a^2)/(k^2/a^2 + m_s^2)$. For $F = \mathrm{d}f/\mathrm{d}R \simeq 1$, the quasistatic linear response has the standard form

$$k^2\Psi = -4\pi G a^2 \bar{\rho}\delta\left(1 + \frac{s}{3}\right), \tag{20}$$

$$k^2\Phi = -4\pi G a^2 \bar{\rho}\delta\left(1 - \frac{s}{3}\right). \tag{21}$$

Thus, with the standard numerator convention,

$$\eta_{\rm std} = \frac{\Phi}{\Psi} = \frac{1 - s/3}{1 + s/3} \longrightarrow \frac{1}{2} \quad (s \to 1), \tag{22}$$

while the reciprocal convention tends to 2. The Weyl/lensing combination remains GR-like, apart from the usual $1/F$ factor:

$$\Sigma \simeq \frac{1}{F} \simeq 1. \tag{23}$$

Consequently, $\eta = 5/4$ and $\Sigma = 9/8$ are not predictions of standard metric $f(R)$ gravity and all forecasts based on those values are withdrawn. The 4/3 enhancement applies to the force potential Ψ, not to a universal free-energy coefficient [13, 14].

Strict convexity of a fixed, environment-independent scalar function $\Phi_{\rm red}(\rho)$ excludes a second minimum of that same function. It does not rule out chameleon screening, because the effective scalar potential itself changes with ambient density. Likewise, changing only a positive-definite metric on configuration space cannot move the minimiser of a fixed scalar function: $\mathrm{grad}_g \Phi = 0$ if and only if $\mathrm{d}\Phi = 0$.

6.2 Source-bound local gate

The 2 July 2026 environment ledger uses

$$|f_R(\mathrm{env})| = |f_{R0}| \left(\frac{R_0}{R_{\rm env}}\right)^{n+1} \tag{24}$$

together with

$$\epsilon_{\rm th} = \frac{|f_{R,\rm env} - f_{R,\rm in}|}{2\Phi_\odot}, \qquad f_{R,\rm in} \simeq 0, \quad \Phi_\odot = 2.12 \times 10^{-6}.$$

The Cassini threshold $\epsilon_{\rm th} < 2.3 \times 10^{-5}$ requires

$$\frac{|f_R(\mathrm{env})|}{|f_{R0}|} < \frac{2\Phi_\odot(2.3 \times 10^{-5})}{|f_{R0}|}. \tag{25}$$

For $n = 1$, this gives

$$\frac{R_{\rm env}}{R_0} \geq 350.787 \quad \text{for } |f_{R0}| = 1.2 \times 10^{-5}, \qquad \frac{R_{\rm env}}{R_0} \geq 101.264 \quad \text{for } |f_{R0}| = 10^{-6}.$$

These threshold calculations are arithmetically correct. No independent local density/curvature model currently supplies the required $R_{\rm env}/R_0$; the ledger therefore permits no claim upgrade. Neither the exploratory best-fit 1.2×10^{-5} nor the conservative 10^{-6} guide is by itself a Cassini proof. The former is background-proxy compatible only and is not demonstrated Cassini-safe.

The previous exploratory MCMC scan used simplified diagonal summary terms rather than the official Planck/DESI likelihoods and encoded Cassini through a linear hard proxy. Its numerical samples may be retained as a background stress test only; they do not establish model preference, official parameter constraints, or local screening.

7 Proof status and next steps

Item	**Status after the 16 July 2026 audit**
Reduced functional	Proven: existence, uniqueness, strict convexity, and the Lambert-W minimiser for specified $B, C > 0$.
Raw mode sum	Invalid as a dimensionally homogeneous functional; motivation only.
Physical $H_0^2/\ln$ scaling	Open and conditional on explicit RG matching; current arithmetic misses observation by about 290.
Pöschl–Teller dynamics	Hilltop transient; no stable de-Sitter attractor and no derived tanh law.
Metric $f(R)$ slip/lensing	Standard unscreened limits are $\eta_{\mathrm{std}} \to 1/2$ (or reciprocal 2) and $\Sigma \simeq 1/F$, not 5/4 and 9/8.
Local screening	Open: threshold ledger correct, but all numerical passes are source-blocked.
Exploratory MCMC	Background-level proxy only; no official Planck/DESI likelihood and no Cassini-safe classification.

The next decisive calculations are: (i) determine dimensionally explicit, renormalised $B(H)$ and $C(H)$; (ii) replace the hilltop potential or derive a stable scalar history from the full equations; (iii) supply an independent galactic/solar Hu–Sawicki profile whose curvature ratio meets the ledger threshold; and (iv) only then rerun an official cosmological likelihood. No release or claim upgrade is warranted before those gates are closed.

AI Disclosure

In the preparation of this manuscript, AI-assisted tools (Claude, Anthropic) were used in a supporting capacity, particularly for literature research, text structuring, and formulation assistance. The conceptual design, scientific argumentation, and responsibility for the entire content rest solely with the author.

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